

Significant figures



1

a Six

e Four

b Four

f Five

c Two

g Three

d Four

h Five

2

a Two

b Three

c Three

d Four

3 Two

4 Two

5 4549

6

a 2600

e 0.72

b 11 000

f 0.030

c 9200

g 5.0

d 220

h 0.0060

7

a 1770

e 9250

b 462 000

f 540

c 6480

g 0.563

d 21 000

h 0.0736

8

a 801 600 000

e 4.504

b 0.06007

f 1.600

c 11.85

g 90.03

d 2.160

h 0.5987

9 7231.0

10

a

i 1000

ii 1100

b 100

11

a 3499

b 2500

c 999

12 512



Absolute error

13 B

14 7.6

15

a 0.5 kg

b 0.5 g

c \$5

16

a 0.05 mL

b 0.5 mm

c 0.05 g

d 0.5 s

e 0.05 °F

f 0.005 kg

g 0.05 m

h 0.005 °C

17 $\frac{1}{2}$ mL

18 We do not know what place value the number has been rounded to.

19 11.51 m

20

a
i 3

ii 422 m³

b
i 2

ii 2.5 mL

Limits of accuracy

21

a 1 cm

b 90 cm

c 0.5 cm

d

i No

ii No

iii No

iv Yes

22

a 10 kg

b 5 kg

c Lower bound: 65 kg, upper bound: 75 kg

23



24 19.995 m

25 12.05 m

26

a

i 13.455 km

ii 13.445 km

b

i 6.45 mi

ii 6.35 mi

c

i 5.5 yd

ii 4.5 yd

d

i 14.005 kg

ii 13.995 kg

e

i 4884.5 m

ii 4883.5 m

f

i \$52.50

ii \$47.50

27

a Maximum: 175, minimum: 165

b Maximum: 3050, minimum: 2950

28

a 26.25

b 0.6

Applications

29

a 1 mm

b 79.5 mm

30

a Upper bound: 272.6925 m^2 , lower bound: 269.3925 m^2

b Upper bound: 66.2 m, lower bound: 65.8 m

31 31.5725 m^2

32

a 13.85 m/s

b 13.95 m/s

33

a 94.4025 m^2

b 92.2625 m^2